HACKATHON CLUB

FUELING INNOVATION WITH EVERY LINE OF CODE.

SR UNIVERSITY

Topic : " Recovery of Deleted Data from XFS and Btrfs Filesystems "

Domain : Cyber Security

Team ID : T122

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Objective

1. Purpose :

- To develop a comprehensive forensic data recovery solution tailored for modern file systems (XFS and BTRF's), enabling efficient recovery of deleted files along with complete metadata critical for digital investigations.

2. Main goals :

- Design robust algorithms to **recover deleted data** from XFS and Btrfs file systems.
- Support **a wide range of file formats**, including documents, logs, scripts, executables, images, archives, and databases
- **Extract complete metadata** (creation, modification, access, and deletion timestamps) to establish digital timelines and evidence integrity.
- Develop a **user-friendly interface (GUI/CLI)** for streamlined analysis and reporting.
- Ensure **data integrity** with verification and validation mechanisms.

3. Expected outcomes :

- Robust tool for recovering data from XFS and Btrfs.
- Improved event reconstruction and evidence discovery.
- Reliable metadata recovery supporting legal use.

Problem Statement

Modern file systems like **XFS** and **BTRFs**, widely used in Linux environments, make **data recovery difficult** due to their complex structures. In forensic investigations, recovering **deleted files and metadata** (timestamps, file names, etc.) is crucial for building timelines and identifying digital evidence.

Current Challenges :

- Traditional tools are built for **FAT** and **NTFS**, not XFS/Btrfs.
- Advanced features like **journaling** and **copy-on-write** complicate recovery.
- Reconstructing **accurate metadata** is difficult and error-prone.
- Lack of **dedicated forensic tools** for modern file systems.

Key points :

- Deleted files often **remain unrecoverable**.
- **Missing timestamps** disrupt timeline analysis.
- Incomplete recovery **undermines legal admissibility**.

Existing System

•Limited support for **XFS and Btrfs** file system recovery.

•Metadata recovery is often **incomplete or i**Data recovery tools (e.g., TestDisk, PhotoRec, Scalpel) are primarily designed for **FAT, NTFS, and ext4** file systems.

•**naccurate**.

Limitations

- **No deep support** for XFS/Btrfs internal structures.
- **Metadata (timestamps, permissions)** often lost or corrupted.
- Ineffective for **copy-on-write** and **journaling** complexities.
- Lacks **forensic-grade reporting** and timeline reconstruction.

Why a New System Is Needed

- Forensics requires **precise, complete recovery** of both data and metadata.
- Current tools **fail to address** modern file system architecture.
- Investigators need a **dedicated, reliable, and legally sound** solution.

Proposed system

1.Introduction:

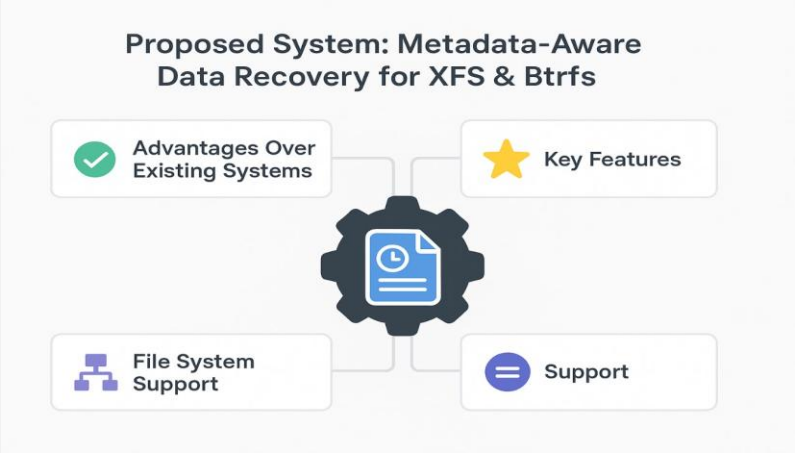
The proposed system is a **specialized forensic recovery tool** designed to extract deleted files and **complete metadata** from **XFS** and **Btrfs** file systems. It addresses the gaps left by traditional tools and is tailored for modern forensic needs

Advantages Over Existing Systems :

- Supports modern Linux file systems (XFS, Btrfs).
- Recovers both files and metadata (MACD: Modification, Access, Creation, Deletion).
- Generates legally sound forensic reports.
- Provides both CLI and GUI interfaces for diverse user preferences.
- Ensures data integrity with built-in verification mechanisms.

Key Features

- **Deep Scan & Recovery**: Identifies deleted data blocks and reassembles file content.
- **Multi-format File Support**: Recovers documents, images, executables, scripts, databases, and more.
- **Complete Metadata Extraction**: Reconstructs access, creation, modification, and deletion timestamps.
- **Forensic Reporting**: Automatically generates evidence-ready reports.
- **Interactive Interface**: Provides an intuitive GUI/CLI for file exploration, preview, and export.



Workflow

Step 01

Identifying deleted data (crime scene investigation)

Step 02

Ensure clear labeling of each step

Step 03

Recovering data fragments (reconstructing a shredded document)

Step 04

Extracting metadata (retrieving security camera footage

Step 05

Provide a brief explanation alongside the diagram

System Architecture

Tech Stack

Languages Used: C, HTML, CSS, JavaScript, etc.

Database Used: Stores recovered file data and metadata.

Database Security: Ensures integrity and confidentiality.

Cloud Service for Hosting: Enables remote access and processing.

APIs Used: Interfaces for data extraction and analysis.

Diagram Analysis: Visual representation of recovery workflow.

Conclusion & Future Scope

Conclusion

The project effectively automated a complex process using modern technologies, demonstrating strong integration of data processing, algorithm design, and user-focused development.

Impact and Benefits

Improved efficiency, minimized errors, and reduced task time. The system is scalable, fast, accurate, and user friendly.

Future Enhancements

Potential upgrades include AI-powered automation, cloud integration, stronger security, cross-platform support, and real-time analytics.

Forward-Thinking Vision

This project sets a solid foundation for future innovations with its modular, scalable design, ready to evolve with technology and user needs.